ARIEL GOODWIN

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EDUCATION

Ph.D. Applied Mathematics 2022 – 2027 (Expected)
Cornell University, Center for Applied Mathematics GPA: 4.00/4.00

Committee: Adrian Lewis (Advisor), Anil Damle, and Soroosh Shafiee

M.Sc. Applied Mathematics 2022 – 2025
Cornell University, Center for Applied Mathematics GPA: 4.00/4.00

B.Sc. Honours Mathematics and Computer Science 2018 – 2022
McGill University, Department of Mathematics GPA: 3.98/4.00

AWARDS & SCHOLARSHIPS

INFORMS Optimization Society Student Paper Prize 2026
Institute for Operations Research and the Management Sciences (INFORMS)

· Awarded for *Stochastic and incremental subgradient methods for convex optimization on Hadamard spaces*; recognizes an outstanding student paper in optimization published in the last 3 years

NSERC Postgraduate Scholarship - Doctoral (PGS D) 2024 – 2027
Natural Sciences and Engineering Research Council of Canada (NSERC), held at Cornell University

· To support doctoral research

Cornell Fellowship 2022
Cornell University

· Recruitment fellowship for academic merit, covering one year of Ph.D. studies

Sir Edward Beatty Memorial Scholarship in Mathematics 2021
McGill University

· For academic achievement in mathematics

NSERC Undergraduate Student Research Award Summer 2021
NSERC, held at McGill University

· Conducted research on convex optimization for statistical estimation, supervised by Rustum Choksi

NSERC Undergraduate Student Research Award Summer 2020
NSERC, held at McGill University

· Conducted research on nonsmooth optimization for projection problems, supervised by Tim Hoheisel

Dr. Feng Qian Scholarship 2020
McGill University

· For academic achievement in computer science

Seba Abbott Smith Scholarship 2018 – 2022
McGill University

· For academic merit and leadership in secondary and postsecondary school

RESEARCH VISITS

Visiting Graduate Student Summer 2026
Halıcıoğlu Data Science Institute, University of California San Diego

· Hosted by Dmitriy Drusvyatskiy for a research project on the geometry of decision trees

NSF Mathematical Sciences Graduate Internship Summer 2023
National Science Foundation (NSF), held at the National Laboratory of the Rockies (formerly NREL)

· Hosted by Jonathan Maack for a research project on manifold optimization for optimal power flow

PUBLICATIONS AND PREPRINTS

A finite set with noncompact closed convex hull in a Hadamard space
Preprint, arXiv:2609.06455, 2026.

Circumcenters and mean sets in Hadamard space: horospherical subgradient methods
With A. S. Lewis, G. López-Acedo, and A. Nicolae. *Mathematics of Operations Research*, 2026.

Minimal enclosing balls via geodesics
With A. S. Lewis. *Preprint*, arXiv:2603.15488, 2026.

Stochastic and incremental subgradient methods for convex optimization on Hadamard spaces
With A. S. Lewis, G. López-Acedo, and A. Nicolae. *Mathematical Programming*, 2026. Recipient of the **INFORMS Optimization Society's 2026 Student Paper Prize.**

Recognizing weighted means in geodesic spaces
With A. S. Lewis, G. López-Acedo, and A. Nicolae. *Foundations of Computational Mathematics*, 2025.

Convex optimization on CAT(0) cubical complexes
With A. S. Lewis, G. López-Acedo, and A. Nicolae. *Advances in Applied Mathematics*, 2025.

Maximum entropy on the mean and the Cramér rate function in statistical estimation and inverse problems: properties, models, and algorithms
With Y. Vaisbourd, R. Choksi, T. Hoheisel, and C.-B. Schönlieb. *Mathematical Programming*, 2025.

Power flow geometry and approximation
With J. Maack and D. Sigler. *IEEE Transactions on Power Systems*, 2025.

Riemannian optimization applied to AC optimal power flow
With J. Maack and D. Sigler. *IEEE Power & Energy Society General Meeting*, 2024.

From perspective maps to epigraphical projections
With M. P. Friedlander and T. Hoheisel. *Mathematics of Operations Research*, 2023.

CONFERENCES AND PRESENTATIONS

Minimum Enclosing Ball Algorithms in Geodesic Spaces — July 2027 (Upcoming)
International Symposium on Mathematical Programming, Amsterdam — Invited Talk

A Cyclic Projections Approach to Minimum Enclosing Balls — Jan. 2027 (Upcoming)
Joint Mathematics Meetings, Chicago — Invited Talk

Minimum Enclosing Ball Algorithms in Geodesic Spaces — Dec. 2026 (Upcoming)
Canadian Mathematical Society Winter Meeting, Montreal — Invited Talk

Subgradient Methods on Nonpositively Curved Geodesic Spaces — Nov. 2026 (Upcoming)
INFORMS Annual Meeting, San Francisco — Prize Talk

Minimum Enclosing Ball Algorithms in Geodesic Spaces — Nov. 2026 (Upcoming)
INFORMS Annual Meeting, San Francisco — Invited Talk

Convex Hulls in Spaces of Nonpositive Curvature — Sept. 2026 (Upcoming)
Geometry Seminar, The Pennsylvania State University — Invited Talk

Minimum Enclosing Ball Algorithms in Geodesic Spaces — Aug. 2026
Institute for Foundations of Data Science Research Expo, University of Wisconsin–Madison — Poster

Subgradient Methods on Nonpositively Curved Geodesic Spaces — June 2026
SIAM Conference on Optimization, University of Edinburgh — Invited Talk

Subgradient Methods in Geodesic Spaces — Oct. 2025
Young Researchers Workshop, Cornell University — Poster

Incremental Minimization in Spaces of Nonpositive Curvature — July 2025
International Conference on Continuous Optimization, University of Southern California — Talk

An Invitation to Hadamard Space — Nov. 2024
Applied Mathematics Student Colloquium, Cornell University — Talk

Incremental Minimization in Spaces of Nonpositive Curvature — Nov. 2024
Midwest Optimization Meeting, University of Waterloo — Poster

The Maximum Entropy on the Mean Method for Linear Inverse Problems — July 2022
International Conference on Continuous Optimization, Lehigh University — Talk

Epigraphical Projections in Nonsmooth Optimization — Aug. 2020
Undergraduate Research Conference, McGill University — Talk

TEACHING EXPERIENCE

Ithaca High School Math Seminar Instructor 2025
Cornell University, Ithaca High School

· Designed and taught a seminar on mathematical optimization for advanced high school students

Teaching Assistant 2023 – 2024
Cornell University

· MATH 2930: Differential Equations for Engineers (Summer 2024)
· MATH 2940: Linear Algebra for Engineers (Spring 2024)
· MATH 1920: Multivariable Calculus for Engineers (Fall 2023)

Teaching Assistant 2019 – 2021
McGill University

· MATH 248: Honours Vector Calculus (Fall 2021)
· PHYS 131: Mechanics and Waves (Fall 2019, 2020, and 2021)

SERVICE AND OUTREACH

Peer Mentor 2023 – Present
Center for Applied Mathematics, Cornell University

· Mentoring applied math Ph.D. students in their transition to graduate school

Undergraduate Research Conference Moderator 2026
Summer Research Conference, University of California San Diego

· Moderated undergraduate research presentations

Undergraduate Mentor 2026
Association for Women in Mathematics, Cornell University

· Mentoring an undergraduate mathematics student

High School Research Mentor 2026
Ithaca High School Math Seminar, Cornell University

· Mentored advanced high school students on a research project in optimization

Directed Reading Program Mentor 2024 – 2025
Cornell University

· Mentored undergraduate students in projects on differential geometry, probability, and optimization

Conference Session Organizer

· *International Symposium on Mathematical Programming (ISMP 2027)*

Reviewer

· *Mathematical Programming, Mathematics of Operations Research, Journal of Optimization Theory and Applications, SIAM Journal on Optimization, SIAM Journal on Applied Algebra and Geometry, IEEE Transactions on Power Systems*

SKILLS AND INTERESTS

Computational Skills	Scientific programming, primarily in Julia and Python, with experience in C++, C, OCaml, Java, and MATLAB; proficient in using AI tools to develop proofs, implement algorithms, and run numerical experiments
Research Interests	Nonsmooth Optimization, Convex Analysis, Non-Euclidean Geometry, Optimization Algorithms, Data Science
Languages	English (Native), French (Intermediate)